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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Francium fluoride
Project / Solution Name	Future Skill AI
Name (Leader)	Chew Yu Hang
Email & Phone (Leader)	zhouyuxing931@gmail.com & +60 16-246 7036
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://future-skill-ai-model.onrender.com/
Code Repository URL (public / access granted to judges)	https://github.com/Yuhang08/Future-Skill-AI-Model

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

FutureSkill AI adopts a modular architecture integrating a user interface, backend processing, AI-driven prediction models, occupational datasets, and cloud hosting.

The frontend, developed using HTML, CSS, and JavaScript, allows users to complete career assessments, explore future job trends, evaluate employability, and access personalised reskilling recommendations.

The backend, built with Python and Flask, processes user inputs and connects the platform's core modules. The Career Matching Model analyses RIASEC personality scores and skill preferences to recommend suitable occupations. The Future Job Trend Model analyses historical labour-market data to estimate future occupational trends. The Employability Predictor evaluates career-related indicators, while the Reskilling Recommendation Module identifies skill gaps and generates personalised learning pathways.

Occupational datasets, including O*NET-derived career information and other labour-market resources support the analytical models. SDG mapping connects occupational tasks with relevant United Nations Sustainable Development Goals.

GitHub manages source code and version control, while Render hosts the Flask application, providing public access to the interactive prototype.

The architecture enables integrated data processing, personalised recommendations, and interactive career exploration.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
CSS3	Website styling and responsive layout	W3C Web Standard
Python (Version to verify)	Backend development, AI modelling and calculations	PSF-2.0
HTML5	Website structure and content	Web Standard (WHATWG/W3C)
JavaScript (ECMAScript)	Frontend interactions and dynamic features	ECMAScript Standard
Flask (≥3.0		

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAI API — GPT-4.1 mini (OpenAI)	Generates personalised skill-gap explanations, reskilling guidance, mock interview questions, and feedback on interview answers.	Paid / Sponsor-provided if accessed through sponsored Replit integration
OpenAI Embeddings API — text-embedding-3-small (OpenAI)	Measures semantic similarity between interview questions to reduce repetition and improve question diversity.	Paid / Sponsor-provided if covered by integration
OpenAI Audio Transcription API — gpt-4o-mini-transcribe (OpenAI)	Converts users' spoken mock interview answers into text for AI-based feedback.	Paid / Sponsor-provided if covered by integration

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Render — Render Services, Inc	Hosts and deploys the FutureSkill AI web application, providing public access to the functional prototype.	Free Tier / Region to verify
GitHub — GitHub, Inc	Source code hosting, version control, team collaboration, and deployment integration.	GitHub Free / Global
Replit — Replit, Inc	Cloud-based development, coding, testing, and prototyping.	Replit Core / Sponsor-provided (US$20 per participant)

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
RIASEC Career Matching Model (v1.0)	Francium Fluoride	Self-developed, rule-based model (not trained)	Apache License 2.0 (Project Repository)
Career Trend Analysis Model (v1.0)	Francium Fluoride	Self-developed, statistical/scenario-based model (not trained)	Apache License 2.0 (Project Repository)
Skill Gap & Reskilling Recommendation Model (v1.0)	Francium Fluoride	Self-developed, rule-based model (not trained)	Apache License 2.0 (Project Repository)
SDG Career Mapping Model (v1.0)	Francium Fluoride	Self-developed, rule-based model (not trained)	Apache License 2.0 (Project Repository)
Employment Scenario & Uncertainty Model	Francium Fluoride	Self-developed, scenario simulation	Apache License 2.0 (Project Repository)
GPT-4.1 mini	OpenAI	Pre-trained	Proprietary / OpenAI API Service Terms
text-embedding-3-small	OpenAI	Pre-trained	Proprietary / OpenAI API Service Terms
gpt-4o-mini-transcribe	OpenAI	Pre-trained	Proprietary / OpenAI API Service Terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET Occupational Interest Profiles (923 Careers)	https://www.onetcenter.org/database.html#individual-files	Public / Processed	CC BY 4.0	N
O*NET 31.0 Task Statements	https://www.onetcenter.org/dictionary/31.0/csv/task_statements.html	Public	CC BY 4.0	N
O*NET 31.0 Career Interest Types	https://www.onetcenter.org/dictionary/31.0/csv/career_interest_types.html	Public	CC BY 4.0	N
O*NET 31.0 Knowledge	https://www.onetcenter.org/dictionary/31.0/csv/knowledge.html	Public	CC BY 4.0	N
O*NET 31.0 Essential Skills	https://www.onetcenter.org/dictionary/31.0/csv/essential_skills.html	Public	CC BY 4.0	N
O*NET 31.0 Transferable Skills	https://www.onetcenter.org/dictionary/31.0/csv/transferable_skills.html	Public	CC BY 4.0	N
O*NET Tasks-to-DWAs	https://www.onetcenter.org/dictionary/31.0/csv/tasks_to_dwas.html	Public	CC BY 4.0	N
O*NET Task Ratings	https://www.onetcenter.org/dictionary/31.0/csv/task_ratings.html	Public	CC BY 4.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Career Category Trends (2021–2030)	https://github.com/Yuhang08/Future-Skill-AI-Model/blob/main/src/career_category_trends.csv	Self-developed / Synthetic	Apache-2.0 (Project)	N
AI Adoption & Development Trends (2021–2030)	https://hai.stanford.edu/ai-index/2026-ai-index-report	Public / Synthetic	Source-specific terms; Apache-2.0 for original project outputs	N
UN Sustainable Development Goals Reference Data	https://sdgs.un.org/goals	Public	UN Terms of Use	N
SDG Mapping Rules & Reference Vocabulary	https://github.com/Yuhang08/Future-Skill-AI-Model	Self-developed / Public-derived	Apache-2.0 for original code; UN terms for source material	N
Reskilling Skills Catalogue	https://github.com/Yuhang08/Future-Skill-AI-Model/blob/main/src/reskilling/catalog.py	Synthetic / Self-developed	Apache-2.0 (Project)	N
RIASEC Assessment Scoring Configuration	https://github.com/Yuhang08/Future-Skill-AI-Model/blob/main/src/assessment_scoring.py	Self-developed	Apache-2.0 (Project)	N
Employment Scenario & Uncertainty Parameters	https://github.com/Yuhang08/Future-Skill-AI-Model	Synthetic / Self-developed	Apache-2.0 (Project)	N
Demo Career Profiles	https://github.com/Yuhang08/Future-Skill-AI-Model	Synthetic / Demo	Apache-2.0 (Project, subject to source rights)	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
User Career Assessment Profiles & Learning Progress	User-submitted questionnaires / Internal SQLite database	Private / Self-collected	User-provided data; applicable privacy terms	Y
Mock Interview Question History	Application-generated session records / Internal SQLite database	Private / Self-collected	Application data; applicable privacy terms	Y

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

/ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT — OpenAI (GPT-6.1 Sol Ultra)	Assisted with Python programming, debugging, mathematical modelling, career matching algorithms, RIASEC calculations, SDG mapping, frontend development, technical explanations, and documentation.	Moderate
Replit AI Agent / Assistant — Replit (version : 2.206.0)	Assisted with website development, code generation, frontend design, debugging, and integration, where used.	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

We used AI outputs as references for brainstorming, planning, and input evaluation, guided by GCCOV approach (Goal, Context, Constraints, Output, Verify). AI helped organise the participant handbook into a five-day plan and task table, which we reviewed and adapted. As non-coders, we used Replit AI with instructions to explain code changes and impacts, preserve original code, maintain transparency, use separate branches, and avoid fabricated content. We refined prompts with ChatGPT Plus, critically questioned suggestions, and tested outputs against our requirements. Our team retained responsibility by reviewing AI-generated changes before adoption, checking functionality, making final decisions ourselves, and ensuring compliance with the organisers' AI-use rules.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

/ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core — US$20 per participant (sponsor-provided access)	Used for AI-assisted coding, website prototyping, debugging, and frontend/backend development.
SDC Hackathon Organisers	Mentor consultation	Received technical feedback and guidance on AI integration, model development, and project improvement.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

No, the sponsor-provided resources were not essential to the core functionality of FutureSkill AI. Replit Core supported coding, debugging, and development, but these tasks could be performed using alternatives such as PyCharm, GitHub, and ChatGPT. The prototype is deployed on Render and primarily relies on independently developed mathematical models, algorithms, and occupational datasets. Therefore, the system can continue operating without sponsor-provided resources.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

FutureSkill AI collects user-provided career assessment responses, RIASEC personality scores, skill levels, career preferences, and learning progress to generate personalised career recommendations and reskilling pathways. User profiles and learning progress may be stored in a local SQLite database using anonymous session identifiers. The application does not require users to provide their real names or contact details. User information is processed through backend algorithms, with session-based access controls to protect individual records. API credentials are stored in server-side environment variables rather than exposed publicly. Optional AI-powered features may transmit relevant user inputs to external AI providers only with appropriate configuration and consent. User records may remain stored until deleted; a fixed retention period has not yet been established. The team aims to minimise data collection, restrict access, and protect user privacy throughout the prototype's operation.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

FutureSkill AI has several technical limitations, including potential AI hallucinations, dataset bias, prediction uncertainty, and dependence on external APIs. Career recommendations rely on RIASEC assessments and occupational datasets, which may not fully reflect individual abilities, changing labour-market conditions, or regional employment differences. Future employment trends are estimates based on historical data and assumptions rather than guaranteed outcomes. SDG mappings may also contain inaccuracies due to differences in occupational responsibilities. To mitigate these risks, the system uses transparent mathematical algorithms, structured datasets, rule-based validation, and explainable recommendation scores. AI-generated explanations are subject to input and output validation, while rule-based recommendations remain available if external AI services fail. API rate limits and service disruptions may temporarily affect optional AI features. Future improvements will focus on dataset validation, model accuracy, fairness testing, and continuous updates.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

/ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

/ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

/ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

/ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

/ Any use of official SDC sponsor resources has been declared in Section 5.

/ The prototype described here is accessible to judges through the links provided in Section 1.

/ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



25/10/2026